

Example 15: Show that the premises "Everyone in this discrete mathematics class has taken a course in computer science" and "Marla is a student in this class" imply the conclusion "Marla has taken a course in computer science."

Soln: $C(x)$: x has taken a course in this computer science field
 $S(x)$: x is a student

Domain: The set of all students in the discrete maths class.

Premise: 1) $\forall x(S(x) \Rightarrow C(x))$ 2) $S(\text{Marla})$

Conclusion: $C(\text{Marla})$

Step	Reason
1. $\forall x(S(x) \Rightarrow C(x))$	Premise
2. $S(\text{Marla})$	Premise
3. $S(\text{Marla}) \Rightarrow C(\text{Marla})$	Universal instantiation on 1
4. $C(\text{Marla})$	Modus ponens on 2, 3

Example 16: Show that the premises "A student in this class has not read the book," and "Everyone in this class passed the first exam" imply the conclusion "Someone who passed the first exam has not read the book".

Solution: $B(x)$: x has ~~not~~ read the book $S(x)$: x is a student of this class.
 $E(x)$: x has passed the first exam.

Domain: The set of all people

Premise: 1) $\exists x(S(x) \wedge \neg B(x))$ 2) $\forall x(S(x) \Rightarrow E(x))$

Conclusion: $\exists x(E(x) \wedge \neg B(x))$